

```
In [10]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use('ggplot')
#for historical programming data visualization which breaks up graph into you know semantic components

In [7]: df = pd.read_csv(r"C:\Users\shah1\OneDrive\Documents\coaster_name.csv")

In [8]: df.head()

Out[7]:
coaster_name Length Speed Location Status Opening Date Type Manufacturer Height restriction Model ... speed1 speed2 speed1_value speed1_unit speed_mph height_value height_unit height_ft Inversions_clean
0 Switchback Railway 600 ft (180 m) 6 mph (9.7 km/h) Coney Island Removed June 16, 1884 Wood LaMarcus Adna Thompson NaN Lift Packed ... 6 mph 9.7 km/h 6.0 mph 6.0 50.0 ft NaN
1 Flip Flap Railway NaN NaN NaN Cleveland, Ohio, United States Removed 1895 Wood Lina Beecher NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
2 Switchback Railway (Euclid Beach Park) NaN NaN NaN Closed NaN Other NaN NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
3 Loop the Loop (Coney Island) NaN NaN NaN Other Removed 1901 Steel Edwin Prescott NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
4 Loop the Loop (Young's Pier) NaN NaN NaN Other Removed 1901 Steel Edwin Prescott NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
5 rows x 56 columns

In [9]: df.shape
Out[9]: (1087, 56)

In [14]: df.tail()

Out[14]:
coaster_name Length Speed Location Status Opening date Type Manufacturer Height restriction Model ... speed1 speed2 speed1_value speed1_unit speed_mph height_value height_unit height_ft
1082 American Drieler Looping 3,444 ft (1,050 m) 53 mph (85 km/h) Other NaN NaN Steel Anton Schwarzkopf 55 in (140 cm) NaN ... 53 mph 85 km/h 53.0 mph 53.0 111.0 ft Na
1083 Pantheon roller coaster 3,326 ft (1,014 m) 73 mph (117 km/h) Busch Gardens Williamsburg Under construction 2022 Steel - Launched Intamin NaN Blitz Coaster ... 73 mph 117 km/h 73.0 mph 73.0 178.0 ft Na
1084 Tron Lightcycle Power Run 3,169.3 ft (966.0 m) 59.3111 mph (95.4 km/h) Other NaN June 16, 2016 Steel - Launched Vekoma 4121 ft (1252 cm) Motorbike roller coaster ... 59.3 mph 95.4 km/h 59.3 mph 59.3 78.1 ft Na
1085 Tumbler 770 ft (230 m) 34 mph (55 km/h) Kings Dominion Under construction NaN Steel - 4th Dimension - Wing Coaster S&S - Saneel Technologies NaN 4D Free Spin ... 34 mph 55 km/h 34.0 mph 34.0 112.0 ft Na
1086 Wonder Woman Flight of Courage 3,300 ft (1,000 m) 58 mph (93 km/h) Six Flags Magic Mountain Under construction 2022 Steel - Single-rail Rocky Mountain Construction NaN Raptor Custom ... 58 mph 93 km/h 58.0 mph 58.0 131.0 ft Na
5 rows x 56 columns

In [16]: df.dtypes

Out[16]:
coaster_name object
Length object
Speed object
Location object
Status object
Opening date object
Type object
Manufacturer object
Height restriction object
Model object
...
speed1 float64
speed2 float64
speed1_value float64
speed1_unit object
speed_mph float64
height_value float64
height_unit object
height_ft float64
Inversions_clean float64
Gforce_clean float64
dtype: object

In [11]: df.describe()

Out[11]:
count 932.000000 1087.000000 812.000000 812.000000 937.000000 937.000000 965.000000 171.000000 1087.000000 362.000000
mean 134722.00 1994.960201 35.977886 -41.265373 53.850374 48.617289 89.575171 103.996491 1.326887 3.824006
std 2.114073 22.870269 15.513596 92.785227 23.395518 16.678031 136.246444 67.329992 2.030054 0.899998
min 0.000000 1884.000000 -48.261700 -123.035700 5.000000 5.000000 4.000000 13.100000 0.000000 3.400000
25% 0.000000 1989.000000 35.031050 -84.552200 40.000000 37.300000 44.000000 51.800000 0.000000 3.800000
50% 0.000000 2000.000000 40.289800 -76.653600 50.000000 49.700000 79.000000 91.200000 0.000000 4.000000
75% 3.000000 2010.000000 44.799600 2.778100 63.000000 58.000000 113.000000 131.200000 2.000000 4.500000
max 14.000000 2022.000000 63.230900 153.426500 240.000000 149.100000 3937.000000 377.300000 14.000000 12.000000

In [15]: df.columns

Out[15]:
Index(['coaster_name', 'Length', 'Speed', 'Location', 'Status', 'Opening date', 'Type', 'Manufacturer', 'Height restriction', 'Model', 'Height', 'speed1', 'speed2', 'speed1_value', 'speed1_unit', 'speed_mph', 'height_value', 'height_unit', 'height_ft', 'Inversions', 'Gforce'],
      dtype='object')

In [31]: df.info()

class: pandas.core.frame.DataFrame
RangeIndex: 1087 entries, 0 to 1086
Data columns (total 56 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   coaster_name          1087 non-null   object
 1   Length                1087 non-null   object
 2   Speed                1087 non-null   object
 3   Location              1087 non-null   object
 4   Status                1087 non-null   object
 5   Opening date          1087 non-null   object
 6   Type                  1087 non-null   object
 7   Manufacturer           1087 non-null   object
 8   Height restriction     1087 non-null   object
 9   Model                 1087 non-null   object
10  Height                1087 non-null   object
11  Inversions             1087 non-null   object
12  Lift/launch system    1087 non-null   object
13  Cost                  1087 non-null   object
14  Trains                1087 non-null   object
15  Park section          1087 non-null   object
16  Duration              1087 non-null   object
17  Capacity              1087 non-null   object
18  G-force               1087 non-null   object
19  Designer              1087 non-null   object
20  Max vertical angle    1087 non-null   object
21  Drop                  1087 non-null   object
22  Soft opening date     1087 non-null   object
23  Fast Lane available   1087 non-null   object
24  Replaced              1087 non-null   object
25  Track layout         1087 non-null   object
26  Fasttrack available   1087 non-null   object
27  Soft opening date.1   1087 non-null   object
28  Closing date         1087 non-null   object
29  Opened               1087 non-null   object
30  Replaced by          1087 non-null   object
31  Website              1087 non-null   object
32  Flash Pass Available  1087 non-null   object
33  Must transfer from wheelchair 1087 non-null   object
34  Theme                1087 non-null   object
35  Single rider line available 1087 non-null   object
36  Restraint Style      1087 non-null   object
37  Flash Pass available.1 1087 non-null   object
38  Acceleration         1087 non-null   object
39  Restraints           1087 non-null   object
40  Name                 1087 non-null   object
41  Year Introduced       1087 non-null   int64
42  Latitude              1087 non-null   float64
43  Longitude             1087 non-null   float64
44  Type_Main             1087 non-null   object
45  Opening_date_clean    1087 non-null   object
46  Speed1                1087 non-null   float64
47  Speed2                1087 non-null   float64
48  Speed1_value          1087 non-null   float64
49  Speed1_unit           1087 non-null   object
50  Speed_mph             1087 non-null   float64
51  Height_value          1087 non-null   float64
52  Height_unit           1087 non-null   object
53  Height_ft            1087 non-null   float64
54  Inversions_clean      1087 non-null   int64
55  Gforce_clean          1087 non-null   float64
56  Gforce                1087 non-null   float64
dtypes: float64(42), int64(12), object(14)
memory usage: 475.7+ KB

In [37]: #Adda pregration
df.drop(['Opening date'], axis=1)

Out[37]:
coaster_name Length Speed Location Status Type Manufacturer Height restriction Model Height ... speed1 speed2 speed1_value speed1_unit speed_mph height_value height_unit height_ft Inversions Gforce
0 Switchback Railway 600 ft (180 m) 6 mph (9.7 km/h) Coney Island Removed Wood LaMarcus Adna Thompson NaN Lift Packed 50 ft (15 m) ... 6 mph 9.7 km/h 6.0 mph 6.0 50.0 ft NaN
1 Flip Flap Railway NaN NaN NaN Sea Lion Park Removed Wood Lina Beecher NaN NaN NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
2 Switchback Railway (Euclid Beach Park) NaN NaN NaN Cleveland, Ohio, United States Closed Other NaN NaN NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
3 Loop the Loop (Coney Island) NaN NaN NaN Other Removed Steel Edwin Prescott NaN NaN NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
4 Loop the Loop (Young's Pier) NaN NaN NaN Other Removed Steel Edwin Prescott NaN NaN NaN NaN ... NaN NaN NaN NaN NaN NaN NaN NaN
...
1082 American Drieler Looping 3,444 ft (1,050 m) 53 mph (85 km/h) Other NaN Steel Anton Schwarzkopf 55 in (140 cm) NaN 111 ft (34 m) ... 53 mph 85 km/h 53.0 mph 53.0 111.0 ft NaN
1083 Pantheon roller coaster 3,326 ft (1,014 m) 73 mph (117 km/h) Busch Gardens Williamsburg Under construction Steel - Launched Intamin NaN Blitz Coaster 178 ft (54 m) ... 73 mph 117 km/h 73.0 mph 73.0 178.0 ft NaN
1084 Tron Lightcycle Power Run 3,169.3 ft (966.0 m) 59.3111 mph (95.4 km/h) Other NaN Steel - Launched Vekoma 4121 ft (1252 cm) Motorbike roller coaster 78.1 ft (23.8 m) ... 59.3 mph 95.4 km/h 59.3 mph 59.3 78.1 ft NaN
1085 Tumbler 770 ft (230 m) 34 mph (55 km/h) Kings Dominion Under construction Steel - 4th Dimension - Wing Coaster S&S - Saneel Technologies NaN 4D Free Spin 112 ft (34 m) ... 34 mph 55 km/h 34.0 mph 34.0 112.0 ft NaN
1086 Wonder Woman Flight of Courage 3,300 ft (1,000 m) 58 mph (93 km/h) Six Flags Magic Mountain Under construction Steel - Single-rail Rocky Mountain Construction NaN Raptor Custom 131 ft (40 m) ... 58 mph 93 km/h 58.0 mph 58.0 131.0 ft NaN
1087 rows x 55 columns

In [31]: df = df[['coaster_name', 'Location', 'Status', 'Manufacturer', 'Year Introduced', 'Latitude', 'Longitude', 'Type_Main', 'Opening_date_clean', 'Speed_mph', 'Height_ft', 'Inversions', 'Gforce']]

In [26]: df['Opening_date_clean'] = pd.to_datetime(df['Opening_date_clean'])

In [27]: df = df.rename(columns={'coaster_name': 'Coaster_Name', 'Year Introduced': 'Year Introduced', 'Opening_date_clean': 'Opening_Date', 'Speed_mph': 'Speed_mph', 'Height_ft': 'Height_ft', 'Inversions': 'Inversions', 'Gforce': 'Gforce'})

In [28]: df.head(5)

Out[28]:
Coaster_Name Location Status Manufacturer Year Introduced latitude longitude Type_Main Opening_Date Speed_mph Height_ft Inversions Gforce
0 Switchback Railway Coney Island Removed LaMarcus Adna Thompson 1884 40.5740 -73.9780 Wood 1884-06-16 6.0 NaN 0 2.0
1 Flip Flap Railway Sea Lion Park Removed Lina Beecher 1895 40.5780 -73.9790 Wood 1895-01-01 NaN NaN 1 12.0
2 Switchback Railway (Euclid Beach Park) Cleveland, Ohio, United States Closed NaN 1896 41.5800 -81.5700 Other NaN NaN NaN 0 NaN
3 Loop the Loop (Coney Island) Other Removed Edwin Prescott 1901 40.5745 -73.9780 Steel 1901-01-01 NaN NaN 0 NaN
4 Loop the Loop (Young's Pier) Other Removed Edwin Prescott 1901 39.3538 -74.4342 Steel 1901-01-01 NaN NaN 1 NaN

In [29]: df.isna().sum()

#Isna function dso function returns a boolean value true if the value is nan and the sum function returns sum of the true value

Out[29]:
Coaster_Name 8
Location 213
Status 8
Manufacturer 59
Year Introduced 8
Latitude 275
Longitude 275
Type_Main 8
Opening_Date 208
Speed_mph 158
Height_ft 916
Inversions 8
Gforce 725
dtype: int64

In [30]: df.loc[df.duplicated()]

Out[30]:
Coaster_Name Location Status Manufacturer Year Introduced latitude longitude Type_Main Opening_Date Speed_mph Height_ft Inversions Gforce

In [31]: df.loc[df.duplicated(subset=['Coaster_Name'])].head(5)

Out[31]:
Coaster_Name Location Status Manufacturer Year Introduced latitude longitude Type_Main Opening_Date Speed_mph Height_ft Inversions Gforce
43 Crystal Beach Cyclone Crystal Beach Park Removed Traver Engineering 1927 42.8617 -79.0598 Wood 1926-01-01 60.0 NaN 0 4.0
60 Derby Racer Revere Beach Removed Fred W. Pearce 1937 42.4200 -70.9860 Wood 1931-01-01 NaN NaN 0 NaN
61 Blue Streak (Conneaut Lake) Conneaut Lake Park Closed NaN 1938 41.6349 -80.3180 Wood 1938-05-23 50.0 NaN 0 NaN
167 Big Thunder Mountain Railroad Other NaN Arrow Development (California and Florida)Dyna... 1980 NaN NaN Steel NaN NaN 0 NaN
237 Thunder Run (Canada's Wonderland) Canada's Wonderland Operating Mack Rides 1986 43.8427 -79.5423 Steel 1981-05-23 39.8 32.8 0 NaN

In [32]: df.query('Coaster_Name == "Crystal Beach Cyclone").head(5)

Out[32]:
Coaster_Name Location Status Manufacturer Year Introduced latitude longitude Type_Main Opening_Date Speed_mph Height_ft Inversions Gforce
39 Crystal Beach Cyclone Crystal Beach Park Removed Traver Engineering 1927 42.8617 -79.0598 Wood 1926-01-01 60.0 NaN 0 4.0
43 Crystal Beach Cyclone Crystal Beach Park Removed Traver Engineering 1927 42.8617 -79.0598 Wood 1926-01-01 60.0 NaN 0 4.0

In [33]: df.columns

Out[33]:
Index(['Coaster_Name', 'Location', 'Status', 'Manufacturer', 'Year Introduced', 'latitude', 'longitude', 'Type_Main', 'Opening_Date', 'Speed_mph', 'Height_ft', 'Inversions', 'Gforce'],
      dtype='object')

In [39]: df = df.loc[~df.duplicated(subset=['Coaster_Name', 'Location', 'Opening_Date'])].reset_index(drop=True).copy()

#Remove dupl(cates based on selected columns

In [40]: #feature Understanding

Out[40]:
df['Year_Introduced'].value_counts()

Out[44]:
Year_Introduced
1999 46
2000 45
2001 38
2002 29
2003 28
...
1954 1
1959 1
1963 1
1995 1
1884 1
Name: count, Length: 180, dtype: int64

In [47]: ax = df['Year_Introduced'].value_counts().head(10).plot(kind='bar', title='Top 10 Year Coaster')
ax.set_xlabel('Year Introduced')
ax.set_ylabel('Count')

Out[47]: Text(0, 0.5, 'Count')

Top 10 Year Coaster

In [40]: ax = df['Speed_mph'].plot(kind='hist', title = 'Coaster Speed')
ax.set_xlabel('Speed mph')

Out[40]: Text(0.5, 0, 'Speed mph')

Coaster Speed

In [40]: ax = df['Speed_mph'].plot(kind='kde', title = 'Coaster speed')
ax.set_xlabel('Speed')
ax.set_ylabel('Kernel Density Estimation')

Out[40]: Text(0.5, 0, 'Speed')

Coaster speed

In [50]: df.plot(kind='scatter', x='Speed_mph', y='Height_ft', hue='Year_Introduced', title = 'Coaster vs height')
plt.show()

Coaster vs height

In [51]: ax = sns.scatterplot(x='Speed_mph', y='Height_ft', hue='Year_Introduced', data = df)
plt.show()

Coaster speed vs height

In [52]: sns.pairplot(df, vars=['Year_Introduced', 'Speed_mph', 'Height_ft', 'Inversions', 'Gforce'],
plt.show())

In [53]: df_corr = df[['Year_Introduced', 'Speed_mph', 'Height_ft', 'Inversions', 'Gforce']].corr()
df_corr

Out[53]:
Year_Introduced Speed_mph Height_ft Inversions Gforce
Year_Introduced 1.000000 0.171078 0.135413 -0.209887 0.160247
Speed_mph 0.171078 1.000000 0.733999 -0.028705 0.607383
Height_ft 0.135413 0.733999 1.000000 -0.079736 0.466482
Inversions -0.209887 -0.028705 -0.079736 1.000000 0.275991
Gforce 0.160247 0.607383 0.466482 0.275991 1.000000

In [54]: sns.heatmap(df_corr, annot=True)

Out[54]: >
```

